PROJECT PRESENTATION

TripForge

A collaborative trip planning web application

Team Members

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Project description

TripForge is a web app where friends or students plan an entire trip together from one place, instead of juggling chat apps, notes and spreadsheets.

Shared workspace

After creating a trip (name, dates, budget) and inviting friends, everyone sees the same itinerary, Google Maps locations, weather forecast, packing checklist and final trip summary.

Decide together

Members post ideas and vote on them. Polls settle quick choices like where to eat. Every member contributes, with no single admin.

Track and discuss

Expenses are added and each member's share is calculated, with no real payments. A comments section and in-app notifications keep the group in sync.

Problem-solving approach

Group trips are planned across chat apps, notes and spreadsheets. We map each pain point to one feature in a single shared workspace.

Scattered information

One shared workspace holds the itinerary, checklist and summary.

Slow, unfair decisions

Idea voting and polls let every member have a say.

Confusing expenses

Expenses are logged and each member's share is calculated automatically.

Blind planning

Google Maps shows routes and distances. A Weather API shows the forecast.

Technology stack used

React

Frontend. Builds the trip workspace, itinerary, voting and expense screens.

Express

Backend framework. Defines the REST API routes for trips, votes and expenses.

Node.js

Backend runtime. Runs the server logic, such as expense share calculation.

MySQL

Database. Stores users, trips, itineraries, votes, expenses and comments.

Request flow: React sends requests to the Express API on Node.js, which reads and writes MySQL.

Supporting tools: JWT and bcrypt for secure login, Google Maps API, Weather API.

Why this stack?

React over plain JS or jQuery

Reusable components suit screens that update constantly, like votes, expenses and checklists. It has a large ecosystem.

Node and Express over PHP or Django

One language, JavaScript, across frontend and backend. Lightweight REST APIs that are quick to build.

MySQL over MongoDB

Our data is relational: users, trips, members, votes and expenses link together. Joins and constraints keep calculations consistent.

JWT and bcrypt for security

Stateless login that scales easily, with passwords stored as bcrypt hashes, never as plain text.

Proof of commits

Repository: github.com/shadow-MJ/Trip_Forge. 11 commits so far, by Joy & Tonmoy.



